

Test Case 8 – Loan Eligibility & Installment Calculation

Request Data

Declared Monthly Installments: 1500

Declared Net Profit: 180000

Declared Business Revenue: 1500000

Requested Loan Amount: 50000

Requested Tenor: 36 months

Date of Birth: 03/12/1979

Bank Statement Credit (Total Income): 1480000.00

SIMAH Installment Loan Balances: 40000

SIMAH Revolving Credit Limits: 10000

SIMAH Monthly Installments: 2000

Step 1: Maximum Tenor Calculation

$\text{monthsTo70} = 292 \text{ months}$

$\text{riskGradeMaxTenor} = 18 \text{ months}$

$\text{maxTenor} = \text{MIN}(292, 18) = 18 \text{ months}$

Step 2: Effective Revenue

$\text{Effective Revenue} = \text{MIN}(\text{Declared Business Revenue}, \text{Bank Statement Credit})$
 $= \text{MIN}(1500000, 1480000.00) = 1480000.00$

Step 3: Exposure Calculation

$\text{Total Exposure} = \text{SIMAH Installment Loan Balances} + \text{SIMAH Revolving Credit Limits}$
 $= 40000 + 10000 = 50000$

$\text{Exposure Percentage} = (50000 / 1480000.00) \times 100 = 3.38\%$

Step 4: Eligible Loan Amount Calculation

$\text{MaxByRevenue} = \text{Effective Revenue} \times 15\% = 1480000.00 \times 0.15 = 222000.00$

$\text{Available Exposure Limit} = (\text{Effective Revenue} \times 30\%) - \text{Total Exposure}$
 $= (1480000.00 \times 0.30) - 50000 = 394000.00$

$\text{Second Cut Limit} = \text{MIN}(222000.00, 394000.00) = 222000.00$

Max Loan Limit by grade = 125000

Eligible Loan Amount = MIN(125000, 222000.00) = 125000

Step 5: Monthly Net Profit

Bank Statement Credit $\times$ 15% = 1480000.00 $\times$ 0.15 = 222000

min(222,000 , 180,000) = 180,000

Monthly Net Profit = 180,000 $\div$ 12 = 15,000

Step 6: Disposable & Available Disposable Income

Disposable Income = monthlyNetProfit $\times$ 0.5

= 15,000 $\times$ 50% = 7500.00

Monthly Obligations = SIMAH Monthly Installments + (5% $\times$ SIMAH Revolving Credit Limits)

= 2,000 + (10,000 $\times$ 0.05) = 2500.00

Available Disposable Income = disposableIncome - monthlyObligations

= 7500.00 - 2500.00 = 5000.00

Step 7: Installment Calculation

Annual Reducing Rate = 27.3%

Monthly Rate = 27.3 $\div$ 100 $\div$ 12 = 0.0227500000

First-Cut Installment = $P \times r / 1 - (1 + r)^{-n}$

Where

n = eligibleTenor

P = eligibleLoanAmount

After apply all there values to formula

First-Cut Installment = 8540.71

Step 8: Maximum Affordable Installment

MIN(monthlyNetProfit $\times$ 0.25, availableDisposableIncome)

25% of Monthly Net Profit = 15000.00 $\times$ 0.25 = 3750.00

MIN(3750.00, 5000.00) = 3750.00

Maximum Affordable Installment = 3750.00

Step 9: Final Eligibility Decision

if(firstCutInstallment < maxInstallment)

finalEligibleLoanAmount = eligibleLoanAmount;

finalInstallmentAmount = firstCutInstallment; (
else

finalInstallmentAmount = maxInstallment;
finalEligibleLoanAmount = maxInstallment $\times (1 - (1 + \text{monthlyRate})^{-(\text{tenor})}) / \text{monthlyRate}$;

the calculation proceeds to the affordability adjustment.

Final Eligible Loan Amount = 54884.19

Step 10: Calculate Final Loan Amount and EMI based on new eligible amount and Sent to LMS

Final Loan Amount = MIN(Final Eligible Loan Amount, Requested Loan Amount)
= MIN(54884.19, 50000) = 50000

Final EMI = $(P \times \text{monthlyRate}) / (1 - (1 + \text{monthlyRate})^{-(n)})$

monthlyRate = annualReducingRate / 12

n = eligibleTenor

P = finalLoanAmount

After applying these values

Final EMI = 3416.28

Final Outcome:

Risk Grade: D,

Final Eligible Loan Amount: 50000,

Final Installment Amount: 3416.28,

First-Cut EMI: 8540.71,

Max Affordable EMI: 3750.00,

Annual Reducing Rate: 27.3,

Eligible Tenor: 18,

Maximum Tenor: 18,

Exposure %: 3.38,

Monthly Net Profit: 15000.00,

Monthly Disposable Income: 7500.00,

Available Disposable Income: 5000.00